AND and OR Gates
VHDL Simulation Report

Name - 최원창 (Choi Won Chang)

Student ID - 2023032615

Major - Robotics Engineering

1. Objective

The purpose of this assignment is to implement four-input AND and OR gates using VHDL and verify their outputs in Logisim-evolution for all 16 binary input combinations.

이 과제의 목표는 VHDL 로 4 입력 AND 및 OR 게이트를 구현하고, Logisim-evolution 에서 16 가지 이진 입력 조합에 대한 출력을 확인하는 것입니다.

2. VHDL Source Code

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity gate4 is
    port (
        a, b, c, d : in  STD_LOGIC;
        out1       : out STD_LOGIC;
        out2       : out STD_LOGIC
    );
end gate4;

architecture Behavioral of gate4 is
begin
    out1 <= a and b and c and d;
    out2 <= a or b or c or d;
end Behavioral;
```

3. Circuit Design

The entity gate4 has four one-bit inputs (a, b, c, d) and two one-bit outputs. Connect out1 to the AND result pin and out2 to the OR result pin. This combinational circuit requires no clock.

gate4 는 1 비트 입력 a, b, c, d 와 1 비트 출력 두 개를 가집니다. out1 은 AND 결과, out2 는 OR 결과 출력 핀에 연결합니다. 조합논리회로이므로 클록은 필요하지 않습니다.

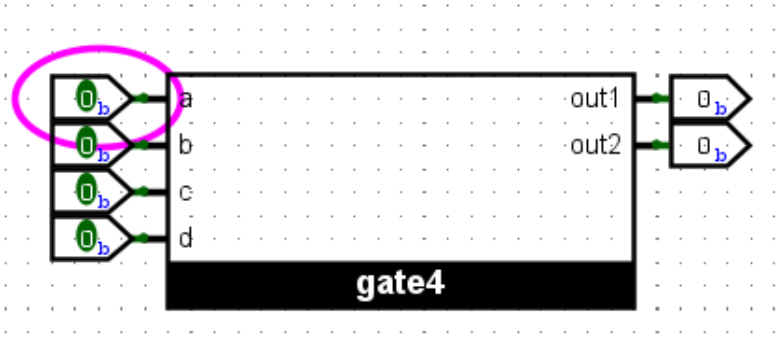


Figure 1. Four-input AND and OR gate circuit in Logisim-evolution.

4. Simulation Results

All 16 binary input combinations were tested in Logisim-evolution. The screenshots show that the AND output is 1 only for input 1111 and the OR output is 0 only for input 0000. Every observed output pair matches the expected truth table.

Logisim-evolution 에서 16 가지 이진 입력 조합을 시험했습니다. 캡처에서 AND 출력은 1111 에서만 1 이고, OR 출력은 0000 에서만 0 입니다. 모든 관측 출력이 예상 진리표와 일치합니다.

Case	a	b	c	d	Expected out1	Expected out2	Observed (out1, out2)
1	0	0	0	0	0	0	(0, 0)
2	0	0	0	1	0	1	(0, 1)
3	0	0	1	0	0	1	(0, 1)
4	0	0	1	1	0	1	(0, 1)
5	0	1	0	0	0	1	(0, 1)
6	0	1	0	1	0	1	(0, 1)
7	0	1	1	0	0	1	(0, 1)
8	0	1	1	1	0	1	(0, 1)
9	1	0	0	0	0	1	(0, 1)
10	1	0	0	1	0	1	(0, 1)
11	1	0	1	0	0	1	(0, 1)
12	1	0	1	1	0	1	(0, 1)
13	1	1	0	0	0	1	(0, 1)
14	1	1	0	1	0	1	(0, 1)
15	1	1	1	0	0	1	(0, 1)
16	1	1	1	1	1	1	(1, 1)

The following figures present the screenshots in ascending a b c d order, from 0000 to 1111. The upper output is out1 (AND), and the lower output is out2 (OR).

다음 그림은 입력 a b c d 를 0000 부터 1111 까지 오름차순으로 정렬한 결과입니다. 위쪽 출력은 out1 (AND), 아래쪽 출력은 out2 (OR)입니다.

4.1 Case 1 a b c d = 0000

Observed outputs: out1 (AND) = 0, out2 (OR) = 0.

관측 출력: out1 (AND) = 0, out2 (OR) = 0. 예상값과 일치합니다.

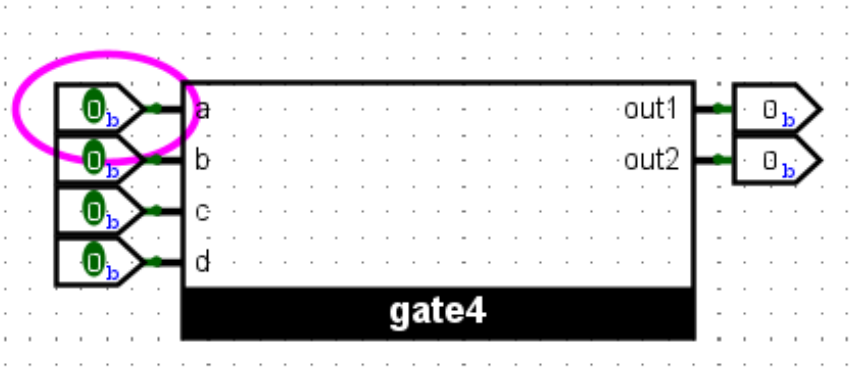


Figure 2. Simulation screenshot for input 0000; observed outputs (0, 0).

4.2 Case 2 a b c d = 0001

Observed outputs: out1 (AND) = 0, out2 (OR) = 1.

관측 출력: out1 (AND) = 0, out2 (OR) = 1. 예상값과 일치합니다.

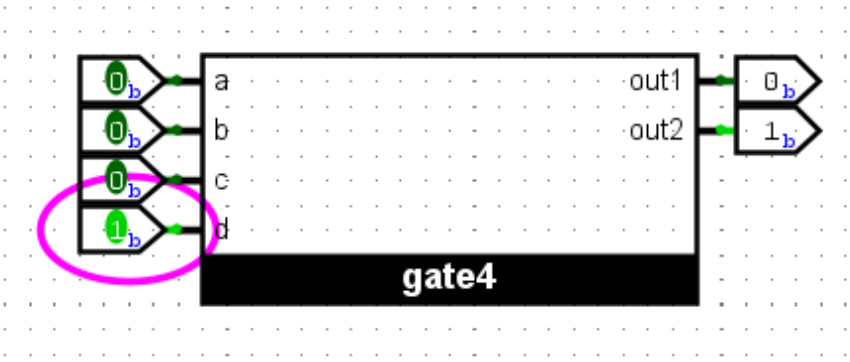


Figure 3. Simulation screenshot for input 0001; observed outputs (0, 1).

4.3 Case 3 a b c d = 0010

Observed outputs: out1 (AND) = 0, out2 (OR) = 1.

관측 출력: out1 (AND) = 0, out2 (OR) = 1. 예상값과 일치합니다.

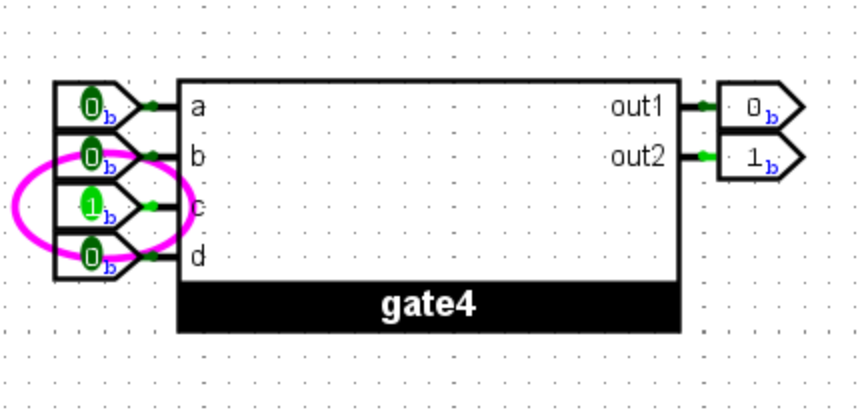


Figure 4. Simulation screenshot for input 0010; observed outputs (0, 1).

4.4 Case 4 a b c d = 0011

Observed outputs: out1 (AND) = 0, out2 (OR) = 1.

관측 출력: out1 (AND) = 0, out2 (OR) = 1. 예상값과 일치합니다.

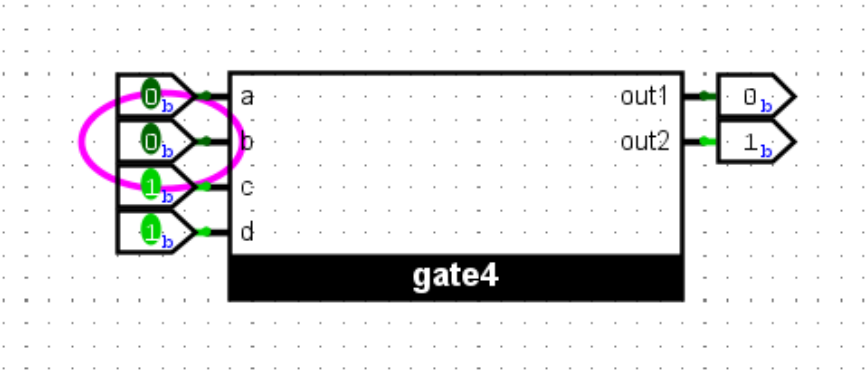


Figure 5. Simulation screenshot for input 0011; observed outputs (0, 1).

4.5 Case 5 a b c d = 0100

Observed outputs: out1 (AND) = 0, out2 (OR) = 1.

관측 출력: out1 (AND) = 0, out2 (OR) = 1. 예상값과 일치합니다.

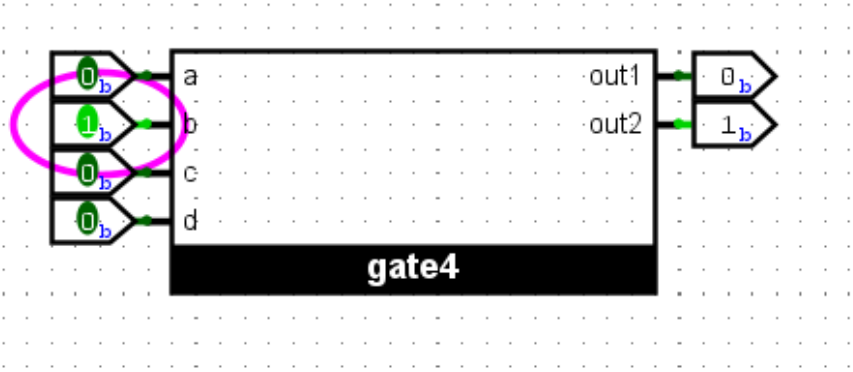


Figure 6. Simulation screenshot for input 0100; observed outputs (0, 1).

4.6 Case 6 a b c d = 0101

Observed outputs: out1 (AND) = 0, out2 (OR) = 1.

관측 출력: out1 (AND) = 0, out2 (OR) = 1. 예상값과 일치합니다.

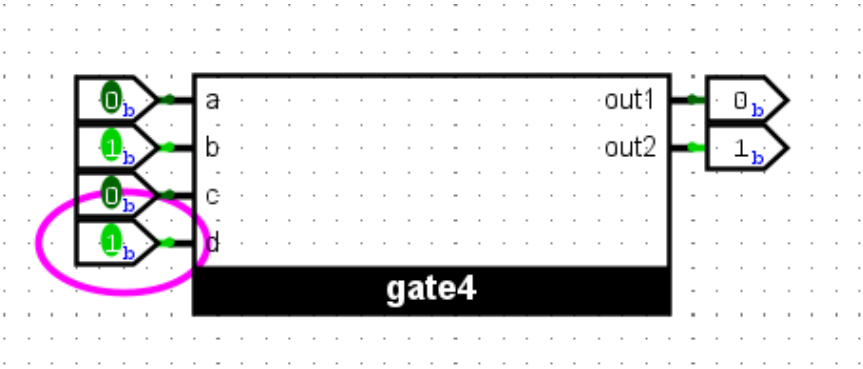


Figure 7. Simulation screenshot for input 0101; observed outputs (0, 1).

4.7 Case 7 a b c d = 0110

Observed outputs: out1 (AND) = 0, out2 (OR) = 1.

관측 출력: out1 (AND) = 0, out2 (OR) = 1. 예상값과 일치합니다.

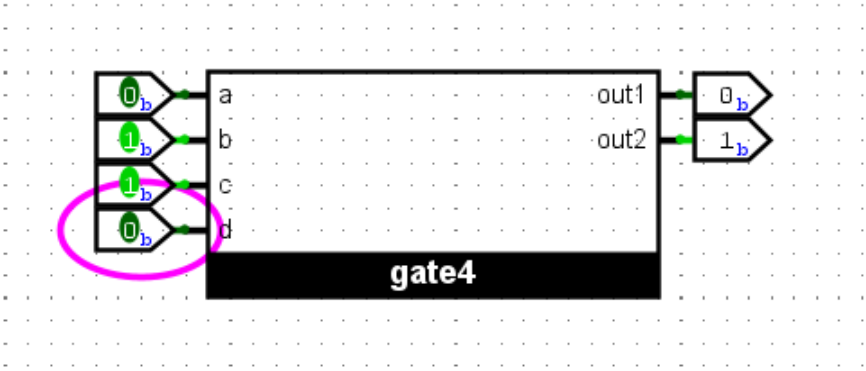


Figure 8. Simulation screenshot for input 0110; observed outputs (0, 1).

4.8 Case 8 a b c d = 0111

Observed outputs: out1 (AND) = 0, out2 (OR) = 1.

관측 출력: out1 (AND) = 0, out2 (OR) = 1. 예상값과 일치합니다.

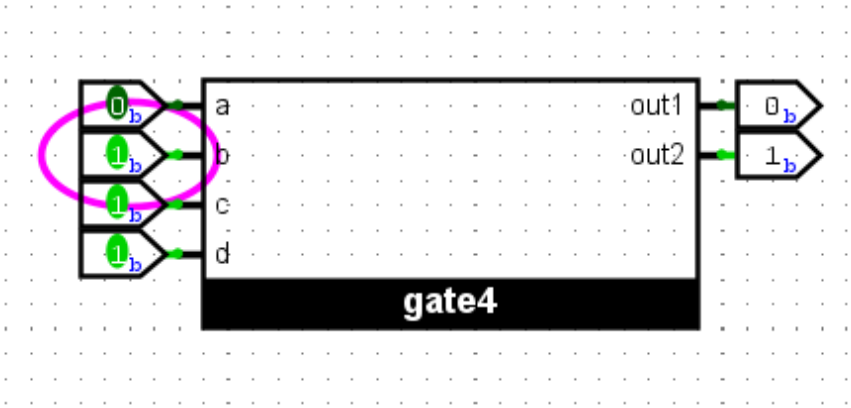


Figure 9. Simulation screenshot for input 0111; observed outputs (0, 1).

4.9 Case 9 a b c d = 1000

Observed outputs: out1 (AND) = 0, out2 (OR) = 1.

관측 출력: out1 (AND) = 0, out2 (OR) = 1. 예상값과 일치합니다.

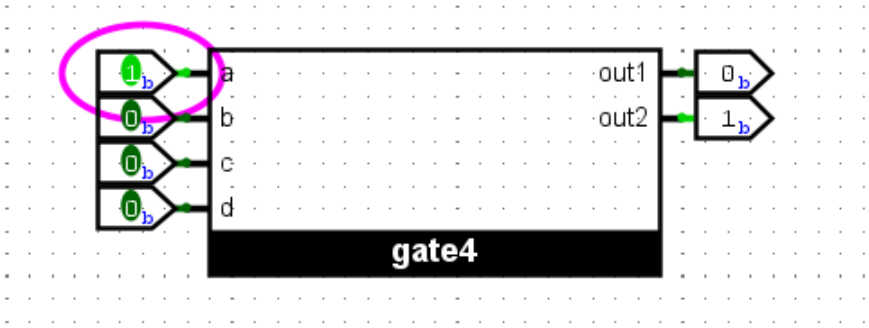


Figure 10. Simulation screenshot for input 1000; observed outputs (0, 1).

4.10 Case 10 a b c d = 1001

Observed outputs: out1 (AND) = 0, out2 (OR) = 1.

관측 출력: out1 (AND) = 0, out2 (OR) = 1. 예상값과 일치합니다.

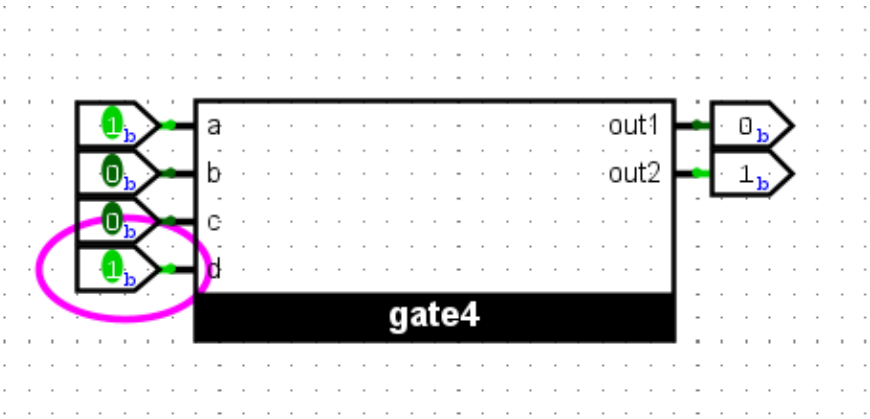


Figure 11. Simulation screenshot for input 1001; observed outputs (0, 1).

4.11 Case 11 a b c d = 1010

Observed outputs: out1 (AND) = 0, out2 (OR) = 1.

관측 출력: out1 (AND) = 0, out2 (OR) = 1. 예상값과 일치합니다.

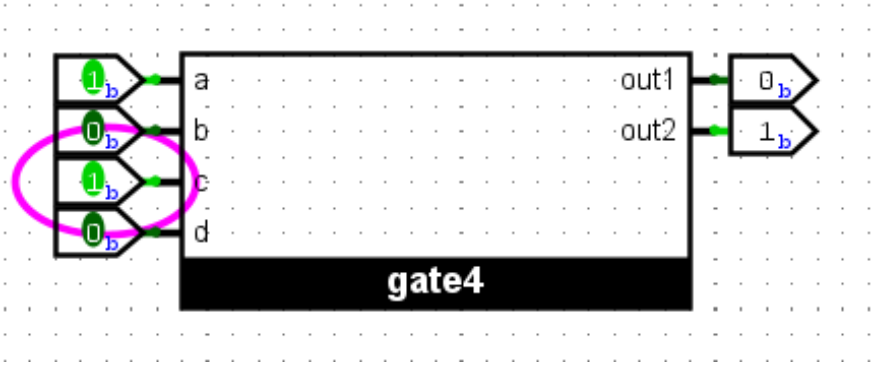


Figure 12. Simulation screenshot for input 1010; observed outputs (0, 1).

4.12 Case 12 a b c d = 1011

Observed outputs: out1 (AND) = 0, out2 (OR) = 1.

관측 출력: out1 (AND) = 0, out2 (OR) = 1. 예상값과 일치합니다.

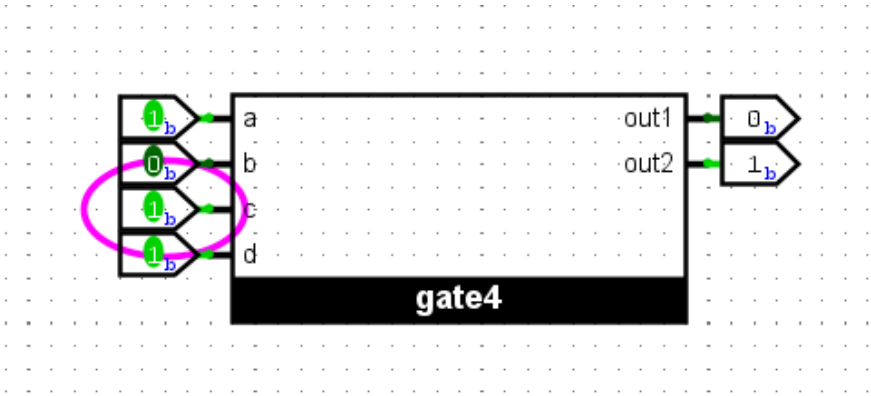


Figure 13. Simulation screenshot for input 1011; observed outputs (0, 1).

4.13 Case 13 a b c d = 1100

Observed outputs: out1 (AND) = 0, out2 (OR) = 1.

관측 출력: out1 (AND) = 0, out2 (OR) = 1. 예상값과 일치합니다.

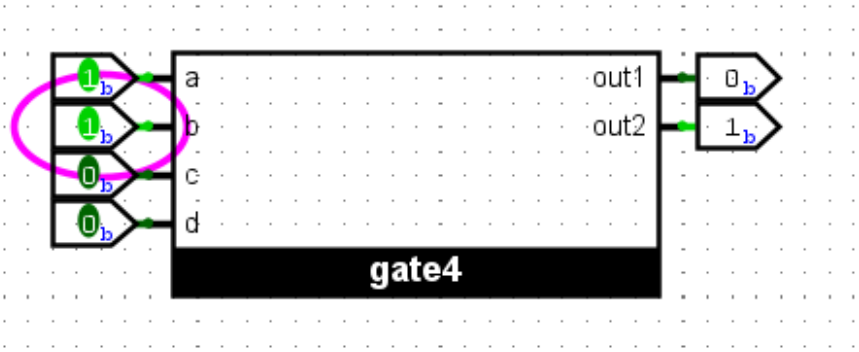


Figure 14. Simulation screenshot for input 1100; observed outputs (0, 1).

4.14 Case 14 a b c d = 1101

Observed outputs: out1 (AND) = 0, out2 (OR) = 1.

관측 출력: out1 (AND) = 0, out2 (OR) = 1. 예상값과 일치합니다.

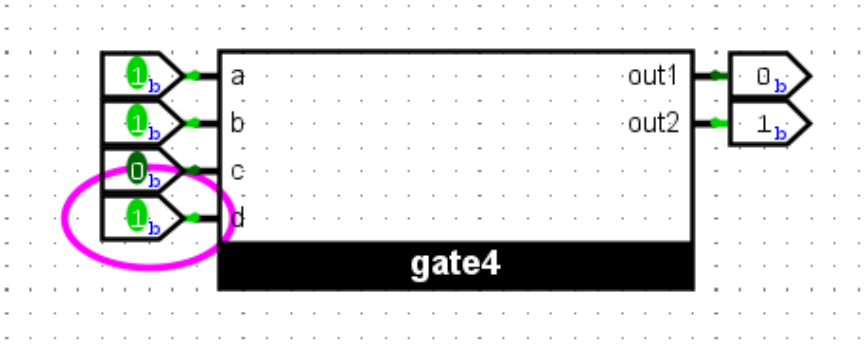


Figure 15. Simulation screenshot for input 1101; observed outputs (0, 1).

4.15 Case 15 a b c d = 1110

Observed outputs: out1 (AND) = 0, out2 (OR) = 1.

관측 출력: out1 (AND) = 0, out2 (OR) = 1. 예상값과 일치합니다.

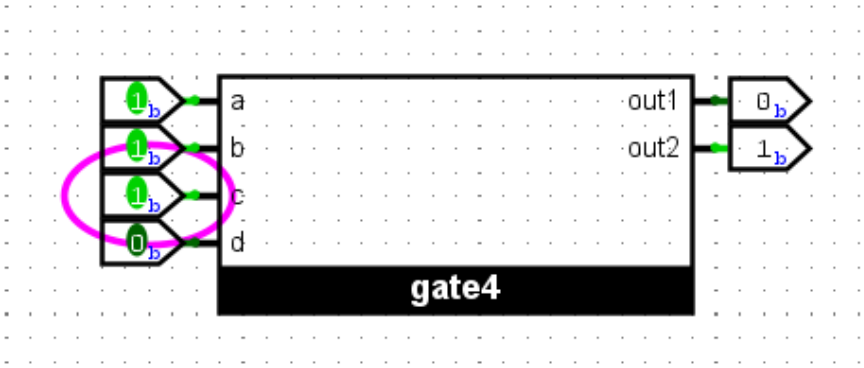


Figure 16. Simulation screenshot for input 1110; observed outputs (0, 1).

4.16 Case 16 a b c d = 1111

Observed outputs: out1 (AND) = 1, out2 (OR) = 1.

관측 출력: out1 (AND) = 1, out2 (OR) = 1. 예상값과 일치합니다.

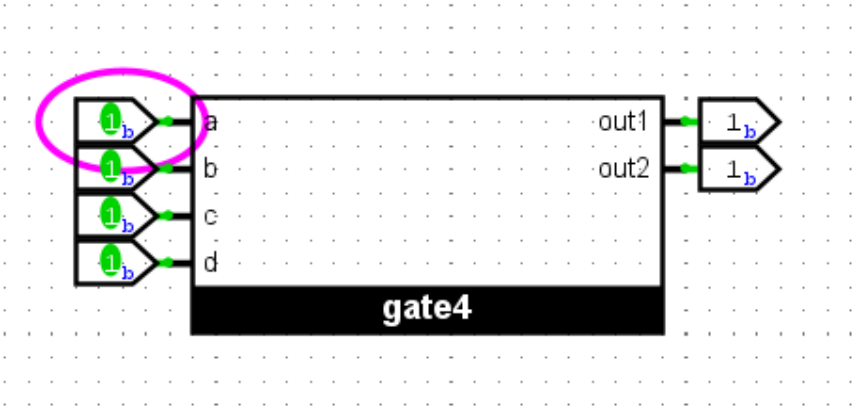


Figure 17. Simulation screenshot for input 1111; observed outputs (1, 1).

5. Conclusion

The four-input AND and OR gates were implemented in VHDL and verified in Logisim-evolution. All 16 screenshots match the truth table: input 0000 produces (0, 0), input 1111 produces (1, 1), and every other input produces (0, 1). Therefore, the circuit correctly performs both required logic operations.

4 입력 AND 및 OR 게이트를 VHDL 로 구현하고 Logisim-evolution 에서 검증했습니다. 16 개 캡처 모두 진리표와 일치합니다. 0000 의 출력은 (0, 0), 1111 은 (1, 1), 나머지는 (0, 1)이므로 회로는 요구된 두 논리 연산을 올바르게 수행합니다.